

Pin-Tzu (Sunny) Lee

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EDUCATION

University of Southern California

Master of Science in Computer Science

GPA: 3.91/4.0

Los Angeles, CA

Aug. 2023 – May 2025

National Chiao Tung University

Bachelor of Science in Computer Science

GPA: 3.84/4.0

Hsinchu, Taiwan

Sep. 2019 – June 2023

EXPERIENCE

Data and Machine Learning Engineer

Jun. 2025 – Present

Northern Lights Post Inc.

Los Angeles, CA

- Architected multi-GPU distributed training pipelines for computer vision models, reducing training time by 60% for large-scale image and video analysis across 9M+ media assets
- Engineered vector search infrastructure using Qdrant database for semantic similarity matching and content recommendation systems
- Developed Django-based backend services and RESTful APIs, integrating ML methodologies with scalable inference endpoints

Machine Learning Engineer Intern

Jan. 2025 – May 2025

Northern Lights Post Inc.

Los Angeles, CA

- Implemented explainable AI (XAI) methodologies utilizing YOLO and SAM2 for object detection and instance segmentation to enhance content interpretability
- Applied computer vision and LLM methodologies to optimize content scoring platform, delivering actionable insights to users
- Leveraged AWS infrastructure (SageMaker, S3) to develop scalable ML pipelines for processing 9M+ images and 10M+ videos

NI LabVIEW Programmer Intern

May 2024 – Aug. 2024

ASML

San Diego, CA

- Integrated new hardware into virtual oscilloscope program, reducing operation time by 95% and improving testing efficiency
- Modularized hardware code using DQMH framework, enhancing code reusability and decreasing code duplication by 40%

PROJECTS

StockTracker: Full-Stack Financial Platform | *MongoDB, Express, Angular, Node.js, Android*

- Architected full-stack financial application using MEAN stack with cross-platform web and Android compatibility
- Engineered real-time data integration with Finnhub and Highcharts APIs for dynamic stock updates and interactive charts

Pyleoclim: Python Package for Paleoclimate Data Analysis | *Python, Machine Learning*

- Contributed to open-source package with 120k+ downloads, implementing LOF algorithm for anomaly detection in time series data
- Authored comprehensive tutorials and established automated testing framework with pytest and continuous integration

TECHNICAL SKILLS

Languages: Python, C/C++, MySQL **Frameworks:** TensorFlow, PyTorch, HuggingFace, Django **Cloud & Databases:** AWS (SageMaker, S3, EC2), Qdrant **Developer Tools:** Git, GitHub

AWARDS

DataFirst Fall 2023: Best Data Science Team Leader **NCTU:** Certificate of Excellence **Hackathon Taiwan Junior 5th:** 1st Place